

Methodology Overview

To assess risk and profitability across the portfolio, I developed two complementary predictive models using XGBoost. The first estimates the likelihood that *a loan application is accepted*. The second estimates the probability that *an issued loan will default*. These two models are linked: the “acceptance-probability” model is used to construct weights that correct for selection bias in the “default-probability” model, so that our default estimates are more representative of the full applicant population, not just the customers we chose to lend to in the past.

When modelling default probability, we can only observe outcomes for loans that were actually approved and issued. Rejected applications never receive a loan, so we never see whether they would have defaulted. If we simply train a default model on accepted loans and apply it to rejected ones, the model is extrapolating its risk estimates to a group it has never seen default, which can systematically bias our predictions on rejected borrowers.

To address this, I first built a model that predicts the probability that a given application would be approved, using all applications in the dataset (both accepted and rejected). The predicted approval probability for each accepted loan is then used to define a weight in the “default-probability” model. Intuitively, accepted loans that “look like” profiles we rarely approve are given a higher weight, because each such loan represents many similar rejected applicants for whom we never observed a default. Conversely, accepted loans from profiles we almost always approve carry less weight. This re-weighting allows the default model to better approximate the risk profile of the whole pipeline of applicants, not just the subset we historically approved.

The final, weighted “default-probability” model is then used to estimate a probability of default for every application in the dataset (accepted and rejected). Using the predicted probability of default for each application, I assigned every applicant to one of four risk bands: (The lowest predicted probability of default in the dataset is 7.2%)

- Band A: PD < 10% (low default risk)
- Band B: 10-20% (moderate default risk)
- Band C: 20-50% (high default risk)
- Band D: ≥ 50% (very high default risk).

For each risk band, a summary report is created, containing fields pertaining to (i) the business profile; (ii) current underwriting behavior; and (iii) risk & return performance.

Insights

- Business Profile

Band	A	B	C	D
% of loans to e-commerce	73.83%	69.89%	72.83%	80.00%
% of loans to SaaS	0.39%	6.91%	6.04%	5.33%
% of loans to other	25.78%	23.20%	21.13%	14.67%
Average business age (months)	34.06	52.37	39.96	26.05
Average cash runway (months)	231.07	185.83	139.45	125.49
% of borrowers with declining revenues	45.70%	34.25%	33.21%	32.00%
% of borrowers with flat revenues	32.42%	40.88%	37.36%	42.67%

% of borrowers with growing revenues	21.88%	24.86%	29.43%	25.33%
Loan-to-monthly-revenue ratio	0.98	1.06	1.31	1.46
Loan-to-cash runway ratio	154.73	993.87	3359.08	3791.56
Average monthly repayment	\$5,992	\$37,434	\$65,880	\$63,044
Repayment-to-revenue ratio	0.182	0.176	0.200	0.236
Note: Bold row name indicates that the metric is computed based on accepted/approved loans only.				

Business Type: Across all bands, the portfolio is heavily skewed towards e-commerce, with SaaS and “Other” business types representing a smaller share of volumes. There is no strong monotonic pattern in sector mix across bands, although the riskiest band (D) is the most concentrated in e-commerce. This suggests that differences in risk across bands are driven more by financial position than by sector mix alone.

Business Age and Cash Runway: In terms of maturity and liquidity, Band D stands out as the youngest group, while Bands A and B contain more established businesses. Cash runway shows an even clearer gradient. Band A has the longest runway by a wide margin, and average runway then declines progressively through Bands B, C and D. In other words, the highest-risk borrowers are typically younger companies with significantly thinner liquidity buffers, whereas lower-risk bands, particularly A and B, are characterised by stronger cash cushions.

Loan Request size: Loan size relative to business capacity also deteriorates across the risk spectrum. The loan-to-revenue ratio rises steadily from just under 1× monthly revenue in Band A to roughly 1.5× in Band D, indicating that loans in the riskiest bands represent a larger claim on current income and are therefore harder to service. The loan-to-cash-runway ratio shows the most dramatic pattern: in Band A, the average loan requested per month of runway is relatively modest, whereas in Bands C and D this ratio is many times higher. Put simply, businesses in the higher-risk bands are taking on loan sizes that are very large relative to the cash runway they have available, leaving little room to absorb shocks.

Monthly repayment: A similar picture emerges when comparing monthly repayment to monthly revenue. Bands A and B have repayment-to-revenue ratios of around 17-18%, while Bands C and D are above 20 percent. This indicates that borrowers in the safer bands have more headroom to meet repayment obligations from ongoing revenues.

Revenue Trends: Revenue trends do vary by band but not in a simple “more risk = more decline” pattern. Band A actually has the highest share of businesses with declining revenues, while B and C has the highest share with increasing revenues. This suggests that, although revenue trend carries information, it is not the primary driver of modelled risk.

- Current underwriting behavior

Band	A	B	C	D
Number of applications	256	362	265	75
Number of loans accepted	158	219	115	46
Acceptance rate	0.617	0.605	0.434	0.613
Fee as a proportion of loan issued	0.076	0.094	0.102	0.100
Note: Bold row name indicates that the metric is computed based on accepted/approved loans only.				

Lending activity: Lending activity is concentrated in the safer segments. Bands A and B together account for the majority of applications (618 out of 958) and approvals (377 out of 538), indicating that Uncapped's GTM efforts are successfully attracting a large volume of lower-risk applicants.

Loan Acceptance rates: However, approval rates are not fully aligned with modelled risk. Acceptance rates in Bands A, B and D are all around 60-61%, whereas Band C is materially lower at about 44%. In effect, Band C is treated more conservatively than the highest-risk band D in terms of approval rate, even though Band D is riskier by construction. This suggests that current underwriting policy does not consistently scale decision stringency with the underlying risk bands.

Loan Fee: Pricing does move in the right direction with risk, but the step-up is modest at the top end. The average fee as a percentage of loan issued rises from around 7.6% in Band A to 9.4% in Band B, and just over 10 percent in Bands C and D. This confirms that Uncapped charges higher fees for riskier bands, which is directionally appropriate. However, the fee differential between Bands C and D is minimal, despite Band D representing the highest-risk borrowers.

- Risk & return performance

Band	A	B	C	D
Number of defaults	0	0	0	38
Default rate on approved loans	0.000	0.000	0.000	0.826
Loss-to-loan issued ratio	0.00	0.00	0.00	0.31
Total \$ return	\$158,476	\$2,384,390	\$2,708,408	-\$4,529,478
% return	3.10%	4.86%	5.67%	-25.78%
Note: Bold row name indicates that the metric is computed based on accepted/approved loans only.				

Defaults and losses: From a risk perspective, all observed credit losses are concentrated in the riskiest-risk band. Bands A, B and C have experienced no defaults to date, whereas Band D has recorded 38 defaults and a realised default rate of approximately 82.6% on accepted loans. This confirms that the model is successfully isolating a genuinely problematic segment. Given this extremely high default rate, a prudent policy would be to avoid approving loans classified in Band D under standard conditions.

% Returns: The return profile mirrors this pattern. On a percentage basis, Bands A-C are all profitable, with positive returns. Band A's return (around 3.1%) is lower than Bands B and C (approximately 4.9% and 5.7% respectively), reflecting the lower fee levels applied to safer loans. Band D, by contrast, is strongly loss-making, with a band-level return of about -25.8%.

\$ Returns: In absolute terms, it is also clear where economic value is created and destroyed. Bands A-C together generate roughly \$5.25 million of profit, with the bulk of this coming from Bands B and C rather than Band A, simply because there are more lendable opportunities in those bands. Band D alone destroys about \$4.53 million, wiping out the majority of the value created by the safer segments.

Recommendations

Current realized % Returns (pre-recommendations)

To assess Uncapped's overall performance against the target of a minimum 2.5% return, I computed the realised global % return across the portfolio,

$$Global \% Return = \frac{\sum_i Return_i}{\sum_i Loan Issued_i} * 100 = 0.60\%$$

This means that, on a realised basis, the portfolio is currently delivering only 0.6%, which is well below the Board's minimum target of 2.5%.

Recommendations

1. Align underwriting decision with risk bands

The analysis of acceptance rates by band indicates that the current underwriting process is not fully aligned with the modelled risk. Given the risk profile, one would generally expect acceptance rates to follow the pattern: Band A > Band B > Band C > Band D. In practice, however, acceptance rates in Bands A, B and D are all around 60-61%, while Band C is materially lower at about 44%. This suggests that the current underwriting rules do not consistently scale decision stringency with underlying credit risk.

A more disciplined approach would be to embed the XGBoost "default-probability" model directly into underwriting. For any new loan request, Uncapped can input the business and loan characteristics into the model to obtain a predicted probability of default. Based on this probability, each application is assigned to one of four risk bands (A-D). These bands can then be used as a simple but powerful decision and pricing framework:

- Band A / Band B (low to moderate risk): generally eligible for approval, subject to standard checks.
- Band C (higher but still manageable risk): approved selectively, depending on Uncapped's risk appetite.
- Band D (very high risk): not approved under standard policy.

2. Focus lending on Bands A and B

The loss analysis shows that every observed default arises from Band D loans, and that the positive returns generated in Bands A-C are almost wiped out by the losses in Band D. On that basis, loans classified in Band D should automatically be declined under the standard policy.

In terms of lending activity and economic contribution, Band B is currently the workhorse of the portfolio. It accounts for a large share of applications, approvals, and positive dollar returns. Band A loans are the safest, but there are fewer such opportunities. If Uncapped were to lend only to Band A, it would leave a material amount of profit on the table, especially as Band B exhibits only low-to-moderate risk.

Whether Band C loans should be approved is a strategic choice. Band C offers higher expected returns but also higher risk than A/B. In this report, I assume a conservative baseline policy in which only Band A and B loans are approved, with Band C treated as an optional "growth" band that could be added later under strict limits and monitoring.

3. Pricing by band

Using the current fee structure by band (as summarised by the fee-to-loan-issued ratio), Bands A and B are priced at approximately 7.6% and 9.4% of loan issued, respectively, with Bands C and D slightly above 10%. Combining these fee levels with the modelled default risk and historical loss patterns, we can estimate expected returns by band.

For each band, the expected \$ return is: (For simplicity, we assume that the loan fee is always fully collected.)

$$E[\$ \text{ Return}] = \text{Fee Charged} - E[\text{Loss from Default}] - \text{Cost of Fund} - \text{Cost of Operations}$$

Where $E[\text{Loss from Default}]$ is estimated as,

$$E[\text{Loss from Default}] = \text{"Default probability"} * \text{"Loan Requested"} * \text{"Loss to loan issued ratio of Band X"}$$

and the expected % return is:

$$E[\% \text{ Return}] = \frac{\text{Total}(E[\$ \text{ Return}])}{\text{Total}(\text{Loan Issued})}$$

Under the current fee structure, the model-implied expected returns by band are:

Band	Expected \$ Returns	Expected % Returns
A	+\$158,476	3.1%
B	+\$2,384,390	4.9%
C	+\$2,708,407	5.7%
D	-\$3,006,844	-17.1%

Limiting approvals to Bands A and B only, while keeping the existing fee structure, is sufficient to lift the overall portfolio above the 2.5% target. In practice, this also creates some headroom to moderately reduce fees in Bands A and/or B if Uncapped wishes to sharpen its pricing and become more competitive, while still remaining above the Board's required return threshold.